

Vishwa Vipulkumar Shah

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EDUCATION

Carnegie Mellon University, School of Computer Science

Master of Science in Intelligent Information Systems (MIS), Language Technologies Institute, CGPA: 4.17

Selected Coursework: Advanced NLP, Multilingual NLP, Multimodal ML, On-Device ML | Research Advisor: [Prof. Maarten Sap](#)

Pittsburgh, PA, USA

Aug 2023 – Dec 2024

Birla Institute of Technology & Science, Pilani

Bachelor of Engineering in Computer Science, CGPA: 9.00/10.00

Goa, India

Aug 2018 – May 2022

EXPERIENCE

Apple

Machine Learning Research Engineer

Seattle, WA

Jan 2025 - Ongoing

- Part of the Input Experience NLP team building language models across input experiences and writing tools for Apple Intelligence.
- Develop and research language models for personalized input experiences and scaling multilingual counterparts.

Datology AI

Research Intern, Supervisor: [Priya Goyal](#)

Redwood City, CA, USA

May 2024 - Aug 2024

- Investigated data curation and augmentation algorithms to improve LLM pretraining efficiency through high-quality inputs.
- Explored data diversification through difficulty and density-based selection in the representational embedding space. Pretraining on this curated data improves training efficiency by 28.6% on large-scale text corpora.
- Synthetically rephrased raw corpora into high-quality learnable formats reducing average perplexity on Pile by 31% and providing 2.24x training speedup compared to equivalent raw-only data.

Meta

AI Resident, Supervisor: [Haoran Li](#)

Menlo Park, CA, USA

July 2022 - July 2023

- Among 35 candidates selected from 2,100 global applicants for AI Residency 2022 cohort to pursue research at Meta AI.
- Project 1: Developed multimodal networks to detect content integrity violations in Virtual Reality worlds, reducing the average time to flag positives by 58%. Observed improvements with diffusion based synthetic generation of visual positives and few-shot prototypical learning for this low-resource task.
- Project 2: Trained Reward models (RM) to induce conversation styles of personalities in LLaMA - [AI Characters](#) through feedback based-learning. Examined improvements with additional preference feedback from AI Judge (RLAIF). The RM achieved 77% alignment with human preference data on tone, style, interestingness axes.

Amazon

Applied Scientist Intern

Bangalore, India

Jan 2022 - Mar 2022

- Part of Moderation Automation and Relevance Systems (MARS) team, which ensures quality advertising with automated detection of defects in video ads. Implemented and fine-tuned 3D ResNet based video models for video quality assessment experiments.
- Trained VGG-13 to detect frames with letter-pillar box defects, resulting in a precision of 93.80% and recall of 62.20%

MIDAS: Multimodal Digital Media Analysis Lab, IIIT Delhi

Research Intern, Supervisor: [Prof. Rajiv Ratn Shah](#) and [Ramit Sawhney](#)

Delhi, India

July 2021 - Dec 2021

- Worked on techniques to improve performance of models across modalities using interpolative data augmentation method - Mixup. Implemented Mixup strategy for Cross-Lingual Zero-shot learning for POS and NLI tasks (*Accepted at MRL, EMNLP*).
- Extended this work to saliency-aware span wise Mixup in the hyperbolic space which outperforms existing methods for low-resource settings across speech and vision (*Accepted at Interspeech*).
- Formulated a set of guidelines for parametrization and training for neural networks to optimally utilize hyperbolic geometry by computing language-specific curvature (*Accepted at NAACL 2024*).

PUBLICATIONS

NormAd: A Framework for Measuring the Cultural Adaptability of Large Language Models

NAACL 2025

Abhinav Rao, Akhila Yerukola, **Vishwa Shah**, Katharina Reinecke, Maarten Sap [[paper link](#)]

2025

Pre-Calc: Learning to Use the Calculator Improves Numeracy in Language Models

AI4Math, ICML 2024

Vishruth Veerendranath, **Vishwa Shah**, Kshitish Ghate [[paper link](#)]

2024

AdaPT: A Set of Guidelines for Hyperbolic Multimodal Multilingual NLP

NAACL Findings

Ramit Sawhney*, Megh Thakkar*, **Vishwa Shah***, Shrey Pandit*, Shafiq Joty [[paper link](#)]

2024

PISA: PolnCare Saliency-Aware Interpolative Augmentation for Low-resource Speech

Interspeech

Ramit Sawhney*, Megh Thakkar*, **Vishwa Shah***, Puneet Mathur, Vasu Sharma, Dinesh Manocha [[paper link](#)] [[code](#)]

2022

Knowledge-based Analogical Reasoning in Neuro-symbolic Latent Spaces

NeSy, IJCLR

Vishwa Shah, Aditya Sharma, Gautam Shroff, Lovekesh Vig, Tirtharaj Dash, Ashwin Srinivasan [[paper link](#)]

2022

Sequence Mixup for Zero-Shot Cross-Lingual Part-Of-Speech Tagging: Extended Abstract

MRL, EMNLP

Megh Thakkar, **Vishwa Shah**, Ramit Sawhney and Debdoot Mukherjee [[program link](#)] [[poster link](#)]

2021

SELECTED PROJECTS

Enhancing Long-term memory in Agents by Learning during Inference

Supervisor: [Prof. Daniel Fried](#) and [Prof. Graham Neubig](#), CMU, LTI

Sep 2024 - Ongoing

- Improving Web-Agents during inference with auto-eval(feedback) to deduce granular sub-trajectories and induce memory.
- Studying failure modes in open-source vs closed-source family of models to tune to specific inabilities

Evaluating Robust use of information in Long Context

Supervisor: [Prof. Graham Neubig](#), CMU, LTI, [\[Report\]](#)

Jan 2024 - Jun 2024

- Analyzing robustness of diverse language models to use relevant information in long-context irrespective of its position by analysing effect of positional encodings, attention, architecture.
- Empirically demonstrated that most long-context adaptation methods(e.g: RoPE, attention based retrieval) are still susceptible to the *lost-in-the-middle* problem
- Also studied this in the context of imminent tasks - RepoEval (repository Level code generation) showing significant drop in function calling performance as context length increases.

Mitigating Hallucinations in Language Models

Supervisor: [Prof. Maarten Sap](#), CMU, LTI, [\[Report\]](#)

Sep 2023 - Jun 2024

- Improving factual correctness in summarization models to mitigate intrinsic hallucination with automatic feedback.
- Developing fine-grained reward models to capture factual contradictions and unsupported spans in the generated summary.

Offensive Hindi-English Tweet Detection, CLSNet Lab, BITS Goa

Supervisor: [Prof. Swati Agarwal](#), BITS Goa

Jun 2021 - May 2022

- This project aims to detect hate speech, trolling, and offensive language in tweets using NLP classification models.
- Built a Federated learning setup to jointly learn from diverse code-switched, code-mixed Hindi-English tweet datasets.
- Built knowledge graphs to link connected concepts in English or Hindi offensive content. This can be used to develop better representations with graph embeddings.

Predicting Stock Price Volatility and Analysing Demographic Bias

[\[Report\]](#) [\[Code\]](#) [\[Paper link\]](#)

Dec 2020 - Jan 2021

- Predicted stock price volatility following earnings calls release on the text-audio paired MAEC dataset.
- Evaluated and demonstrated the existence of Demographic bias in SoTA architectures for volatility prediction. Analysed the input semantic bias and modality wise bias to gain insights (*Work under review*).

SKILLS

Languages: Python, C, C++, MySQL, Shell script, Java

Toolkits: PyTorch, PySpark, Composer (MosaicML), TensorFlow, Keras, xFormers, Flume, Dagster, Flyte

TEACHING EXPERIENCE

- Advanced Natural Language Processing, 11-711**
Teaching Assistant, Carnegie Mellon University, [\[Course website\]](#) Jan 2024 - May 2024
- Natural Language Processing, CSF429**
Teaching Assistant, BITS Pilani, K K Birla Goa Campus, [\[Course website\]](#) Aug 2021 - Dec 2021
- Introduction to Machine Learning and Deep Learning**
Instructor, Center for Technical Education, BITS Pilani, K K Birla Goa Campus Jan 2021 - May 2021
- Discrete Structures for Computer Science, CSF222**
Teaching Assistant, BITS Pilani, K K Birla Goa Campus Sep 2020 - Dec 2020
- Introduction to Computer Programming, CSF111**
Teaching Assistant, BITS Pilani, K K Birla Goa Campus Jan 2020 - May 2020

VOLUNTEERING EXPERIENCE

- Women in Machine Learning and Data Science, Goa chapter**
Campus Representative, Organized events related to Machine Learning for female aspirants June 2020 - May 2022
- BITS Goa Women in Tech**
Mentor, Guided a group of 6 second-year female students across CS topics of their interests June 2020 - Dec 2020
- Peer Mentorship Programme BITS Pilani, Goa Campus**
Mentor, Guided a group of 8 first-year students in academic and co-curricular activities Aug 2019 - May 2020